

“Personal Knowledge Management System”

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Submitted by

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INTRODUCTION

The **Personal Knowledge Management (PKM)** System is a Java-based web application designed to simplify, organize, and enhance the way individuals manage their learning resources and knowledge. In today's fast-paced digital environment, users consume information from multiple sources such as online courses, books, articles, and videos. However, managing this information manually often leads to disorganization, duplication, and difficulty in retrieval.

This project addresses these challenges by providing a structured digital platform where users can efficiently store, categorize, and track their learning materials. Instead of relying on scattered notes or unorganized files, the system allows users to create domains, add subjects under each domain, and attach relevant resources.

The system also includes progress tracking and learning streak features, helping users stay consistent and motivated in their learning journey. The application is developed using **JSP, HTML, CSS, JavaScript, and MySQL**, ensuring a dynamic, responsive, and reliable system for knowledge management.

Overall, the PKM system aims to improve productivity, maintain organization, and provide users with a centralized platform to manage their personal and academic knowledge effectively.

Overview

- The Personal Knowledge Management (PKM) System is a web-based application designed to organize and manage learning resources efficiently.
- The system allows users to create and manage multiple **domains** such as Programming, College, or Personal Development.
- Under each domain, users can add **subjects** to categorize their knowledge in a structured manner.
- Each subject can contain multiple **resources** like videos, articles, books, or notes for better organization.
- The system provides a **dashboard** that displays an overview of total domains, subjects, resources, and learning progress.
- A **tracking module** is implemented to monitor task status such as completed, pending, and in-progress activities.
- The system includes a **learning streak feature** that tracks user consistency based on daily activity.
- Users can easily **search and filter** their subjects and resources for quick access.
- The application is developed using **JSP for backend processing, HTML/CSS for UI design, JavaScript for interactivity, and MySQL for database management**.
- The system ensures **efficient data storage, easy retrieval, and improved productivity** by providing a centralized platform for knowledge management.

AIM & OBJECTIVES

AIM :

The aim of the Personal Knowledge Management (PKM) System is to design and develop an efficient, user-friendly, and structured web-based platform for organizing, managing, and tracking personal learning resources, enhancing productivity, improving knowledge retention, and reducing the complexity of handling scattered information across multiple sources.

OBJECTIVES :

1. To provide user authentication

- Enable secure login functionality using user credentials.
- Maintain session-based access to ensure only authorized users can access the system.

2. To manage domains effectively

- Allow users to create and organize different domains such as Programming, College, or Personal Development.
- Store domain information in the database for structured categorization.

3. To organize subjects under domains

- Enable users to add subjects under each domain.
- Maintain a hierarchical structure for better knowledge organization.

4. To manage learning resources

- Allow users to add resources such as videos, articles, books, and notes under each subject.
- Store resource details including type, name, and links for easy access.

5. To implement progress tracking

- Track the status of tasks as completed, pending, or in-progress.
- Help users monitor their learning progress effectively.

6. To maintain learning streak

- Calculate and display user consistency based on daily completed tasks.
- Encourage regular learning habits.

7. To display dashboard insights

- Provide a dashboard showing total domains, subjects, resources, and task progress.
- Offer a quick overview of user activity.

8. To enable search and filtering

- Allow users to search and filter subjects and resources easily.
- Improve accessibility and usability of stored data.

9. To ensure efficient database management

- Use MySQL to store and retrieve data efficiently.
- Maintain proper relationships between domain, subject, resource, and tracking tables.

10. To design a user-friendly interface

- Develop a clean, responsive, and intuitive UI using HTML, CSS, and JavaScript.
- Enhance user experience and ease of navigation.

SOFTWARE AND HARDWARE

Software Required:

- **Java (JDK 8 or above)**
- **Visual Studio Code**
- **Technologies:** Java Server Pages (JSP), HTML5, CSS3, JavaScript
- **Database:** MySQL
- **Operating System:** Windows 10 / Windows 11 / Linux
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Web Server:** Apache Tomcat Server
- **JSP and Servlets**
- **JDBC (Java Database Connectivity)** for database interaction

Hardware Required:

- Any desktop/laptop with minimum 4GB RAM
- **Processor:** Intel i3 or above
- **RAM:** 4GB minimum (8GB recommended)
- **Storage:** Minimum 500MB free space
- **Input Devices:** Keyboard and Mouse

SYSTEM DESIGN / PROJECT FLOW

The Personal Knowledge Management (PKM) System follows a structured workflow using JSP, sessions, and MySQL database. The system is divided into logical modules to ensure efficient data management, secure access, and smooth user interaction.

Overall Project Flow:

- User opens the Personal Knowledge Management System.
- User logs in using registered credentials.
- System authenticates the user using database records.
- After successful login, the user is redirected to the dashboard.
- User can manage domains, subjects, and resources.
- All data is stored and retrieved from the MySQL database.
- Sessions are used to maintain secure login states.
- User logs out and session is destroyed.

User Module Flow:

- User logs in using username and password
- System validates credentials
- Dashboard is displayed after successful login

Domain Module Flow:

- User adds domain with name and description
- Data is stored in **domain table**
- Domains are displayed on dashboard

Subject Module Flow:

- User adds subject under selected domain
- Subject details stored in **subject table**
- Subjects displayed under respective domains

Resource Module Flow:

- User adds resources under a subject
- Resource details stored in **resource table**
- Resources displayed for each subject

Tracking Module Flow:

- User updates task status (Completed / Pending / In Progress)
- Data stored in **tracking table**
- Progress and learning streak are updated

Data Flow Design:

File Name	Purpose
User	Stores user credentials
Domain	Stores domain details
Subject	Stores subjects under domains
Resource	Stores resources under subjects
Tracking	Stores overall progress, subject wise + domain wise

Session Management Flow:

- Session created after login
- Stores user_id and username
- Validates access on each page
- Logout destroys session

Security Flow:

- User authentication using login credentials
- Hashing MD5 is used to encrypt users' passwords
- Session-based access control
- Data filtered using user_id
- Secure queries using PreparedStatement

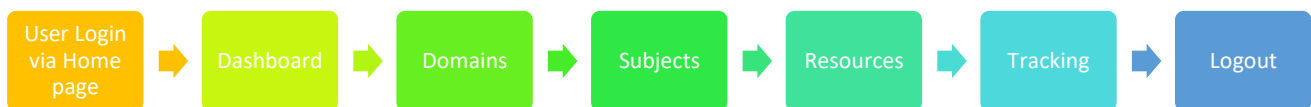
System Flowchart:

Figure1 : System Flowchart

CONCEPT USED

JSP Concepts Used:



Figure2 : JSP Concepts Used.

HTML Concepts Used:

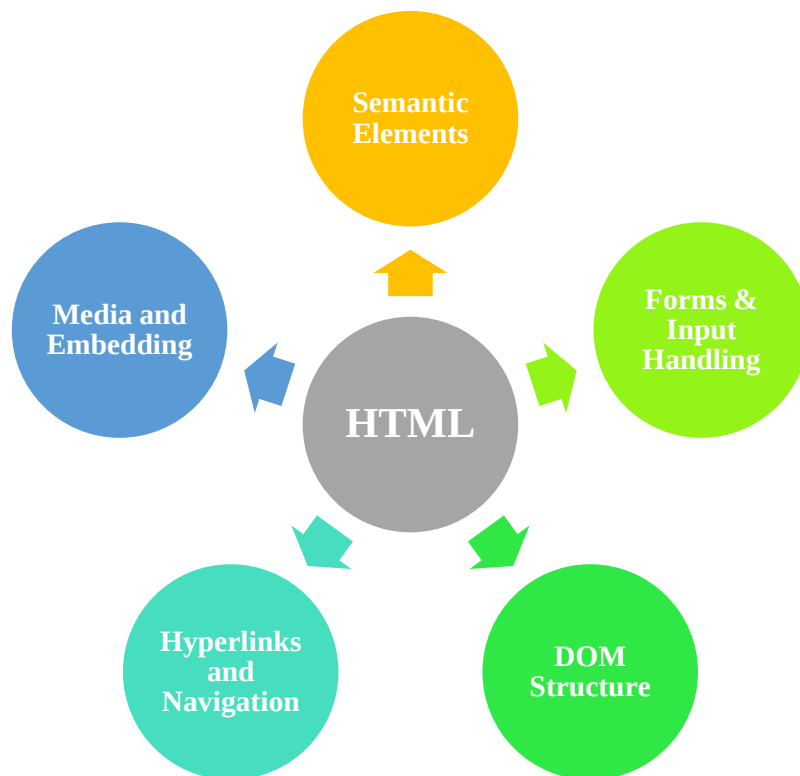


Figure3 : HTML Concepts Used.

CSS Concepts Used:

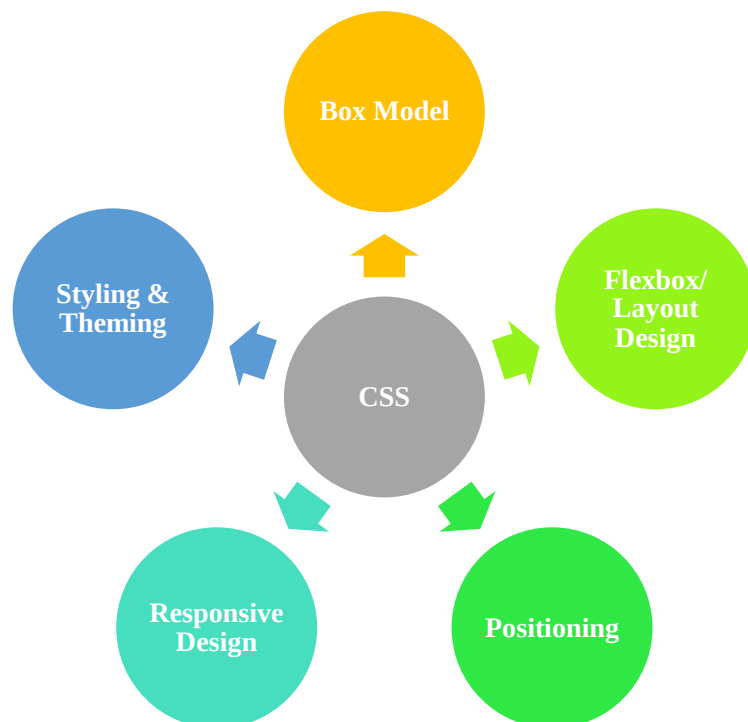


Figure4 : CSS Concepts Used.

OUTPUT SCREENSHOTS

Image1: Home (Login/ Registration) Page

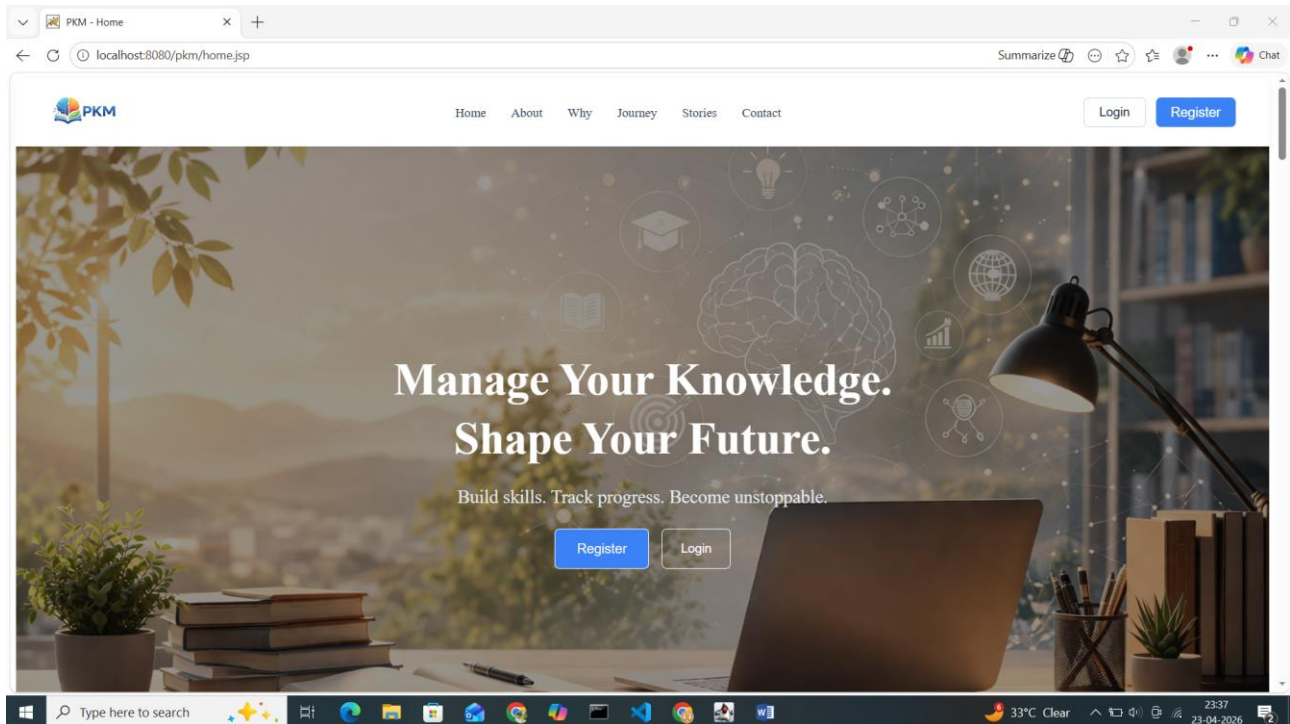


Image2 : Home page continue (About section)

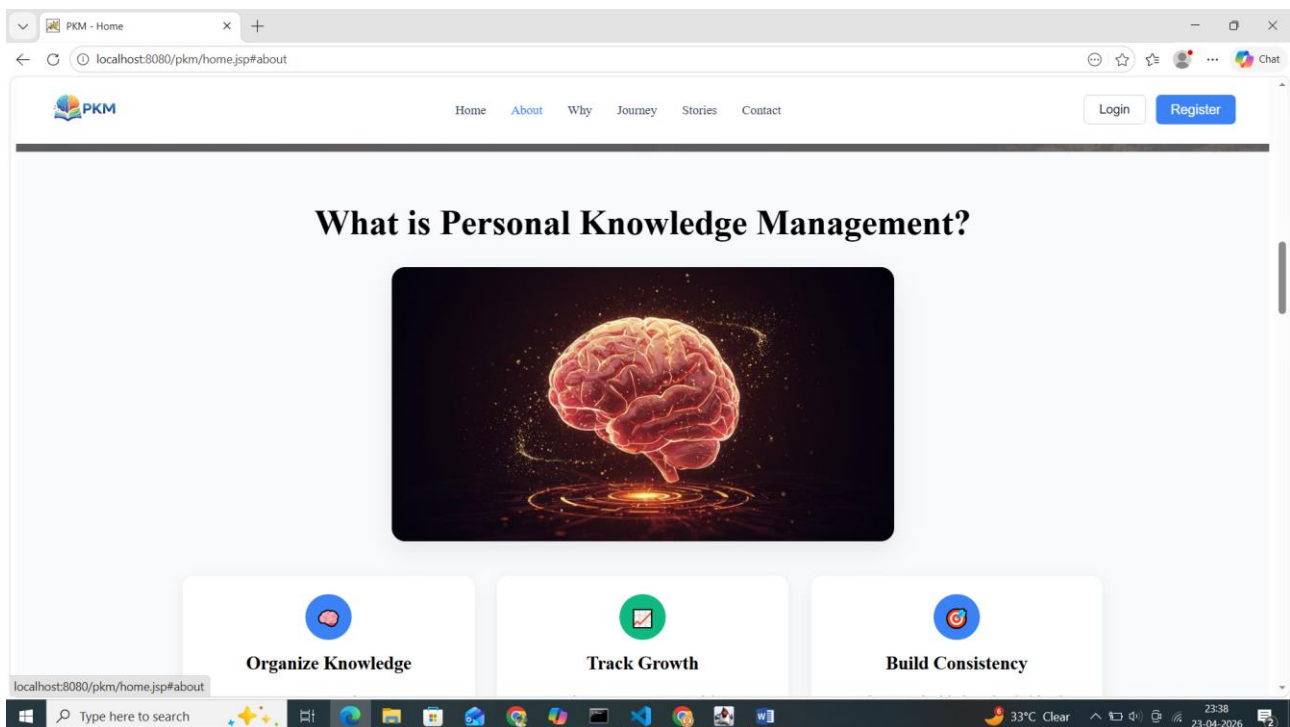


Image3 : Home page continue (Stories Section)

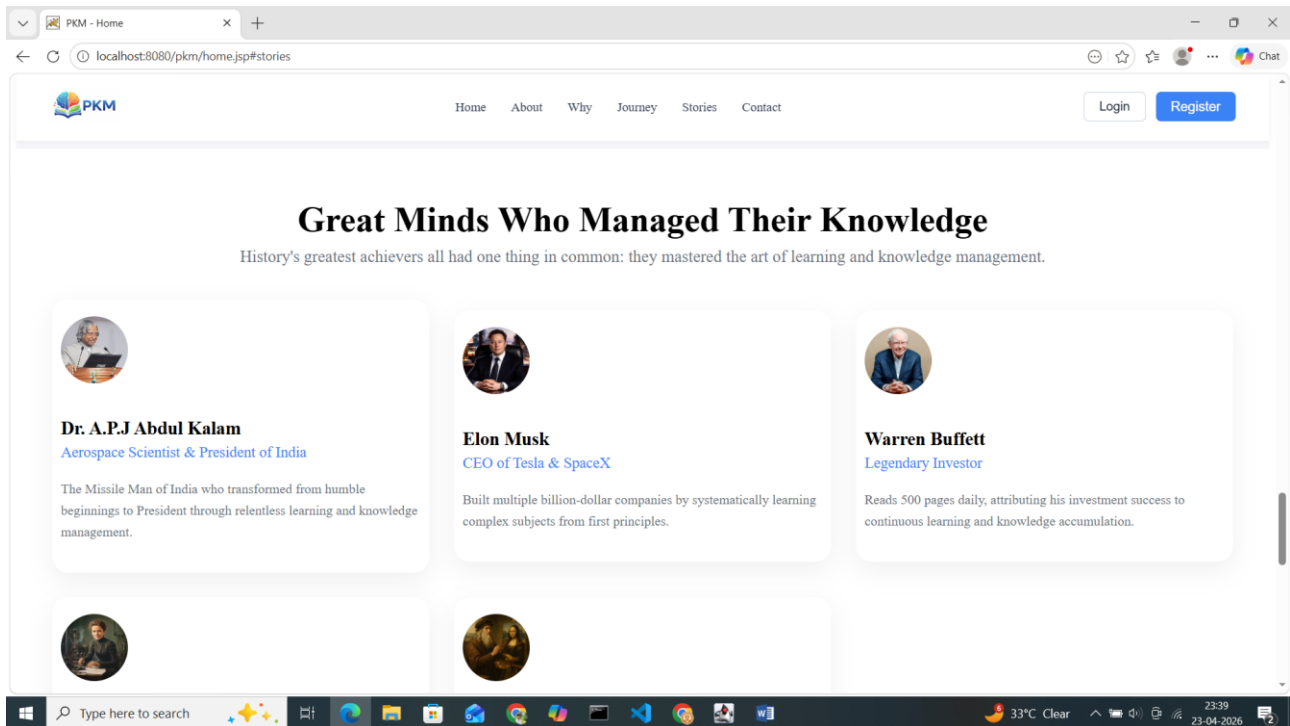


Image4 : Home page continue (Contact Section)

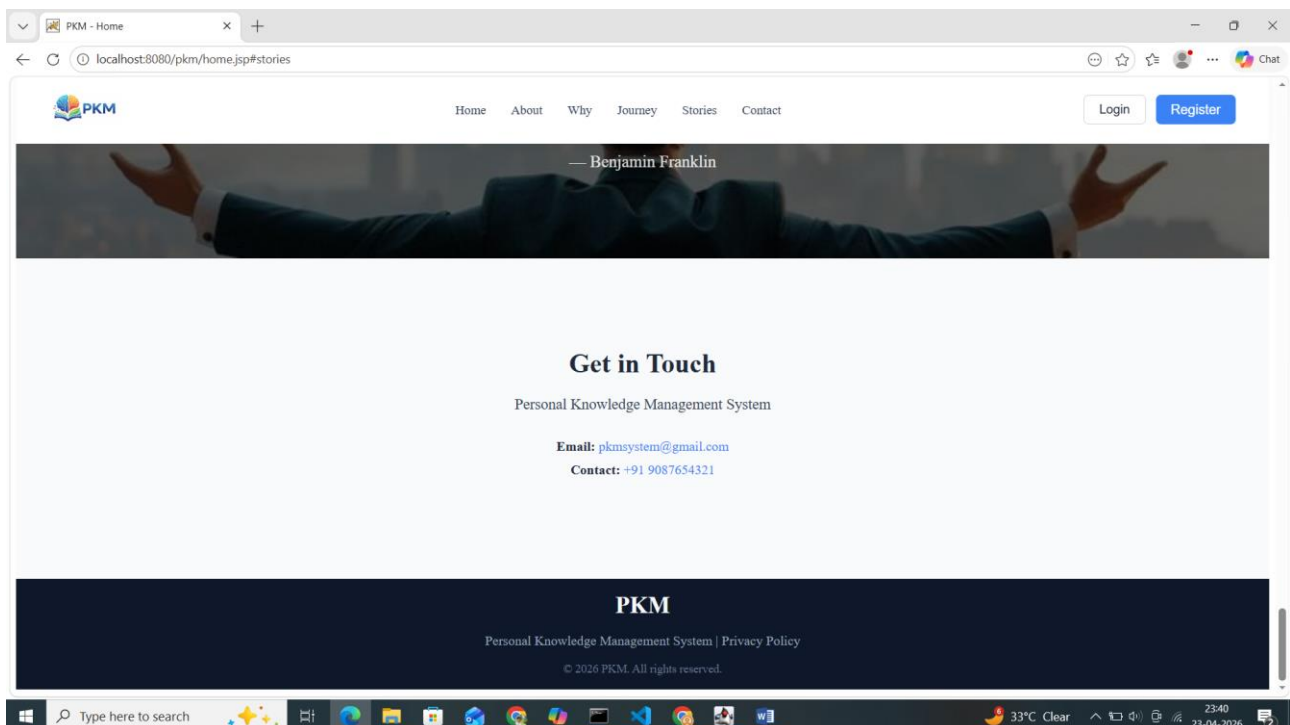


Image5 : Login/Registration form Page

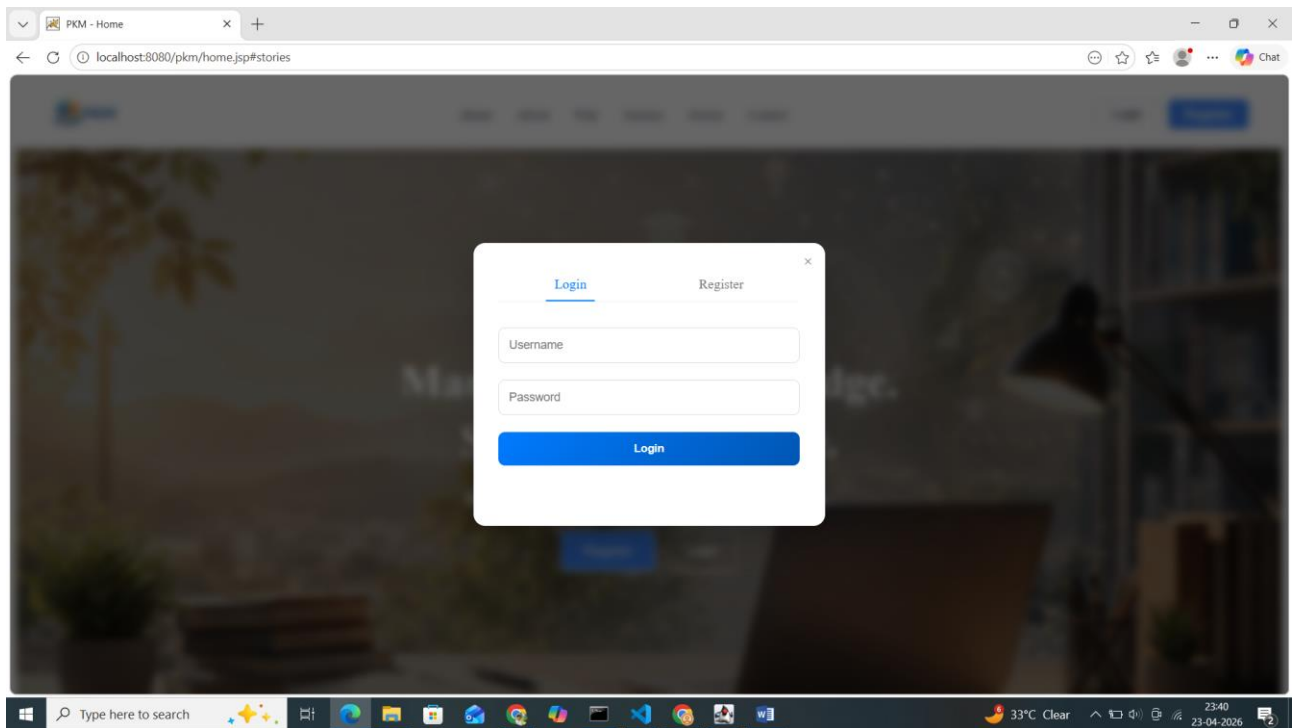


Image6 : Dashboard

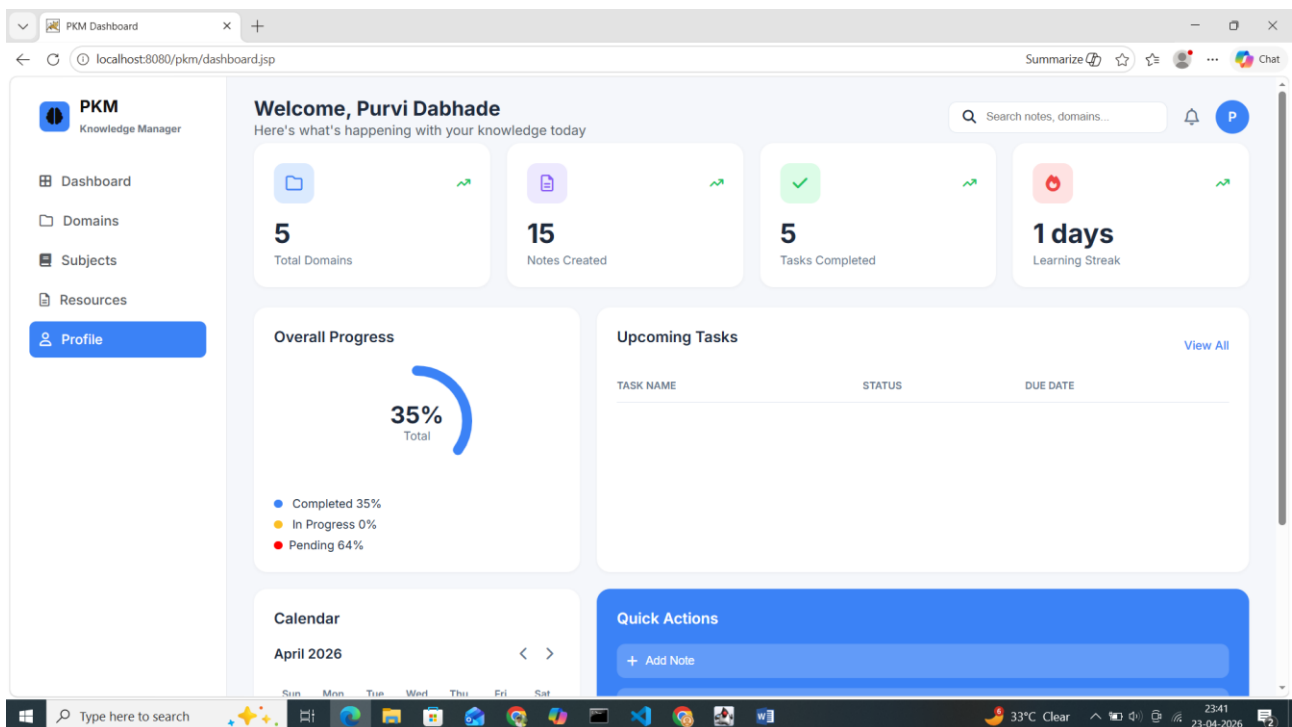


Image7 : Dashboard continue

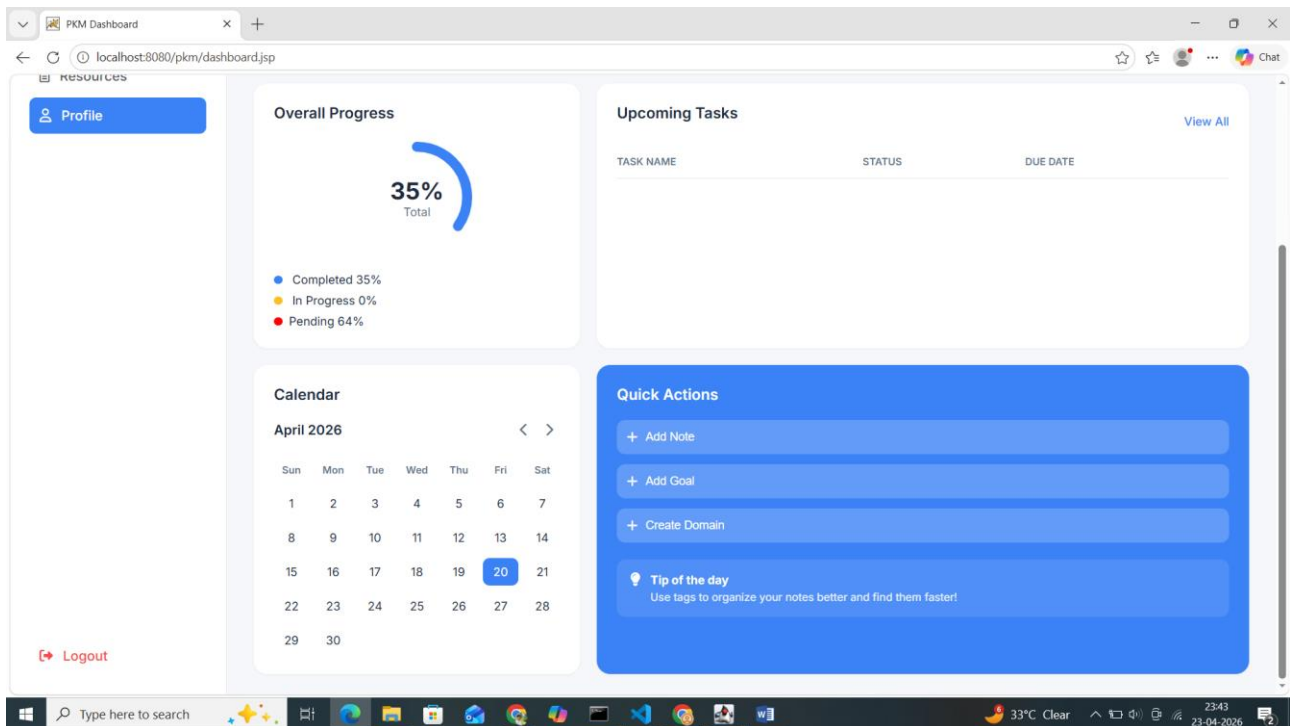


Image8 : Domain Page

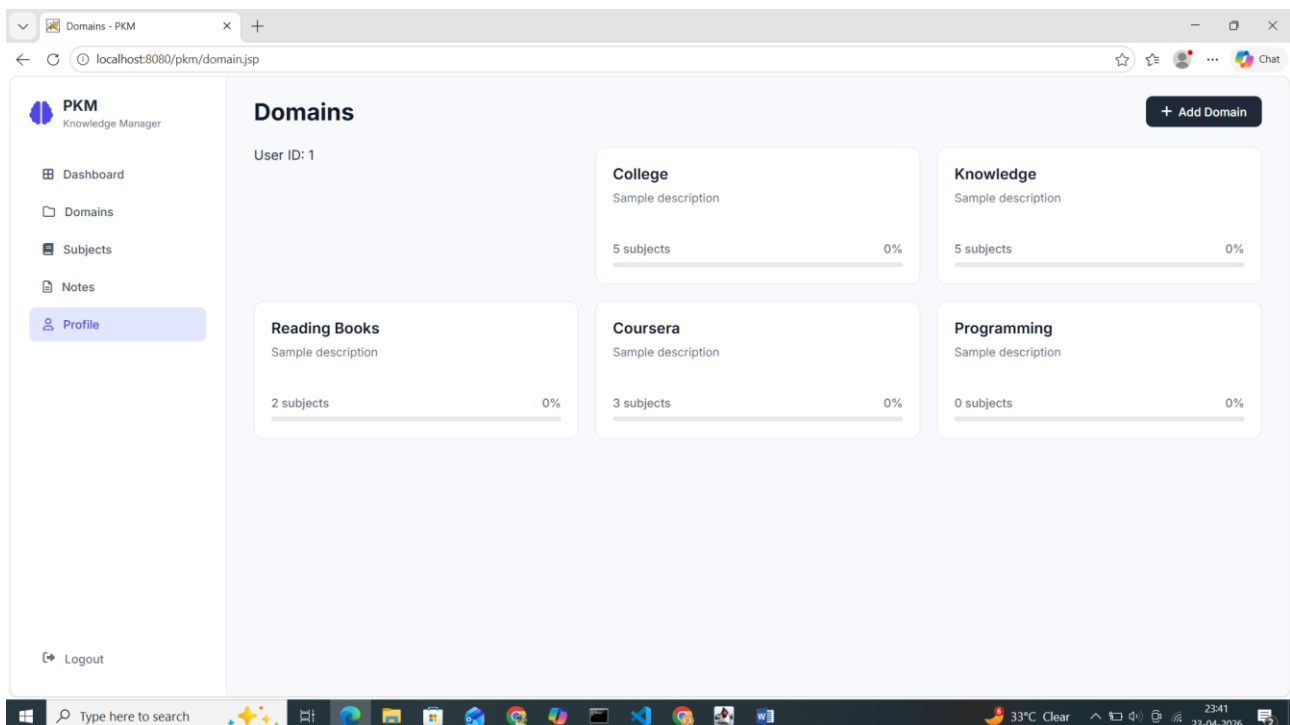


Image9 : Resources Page

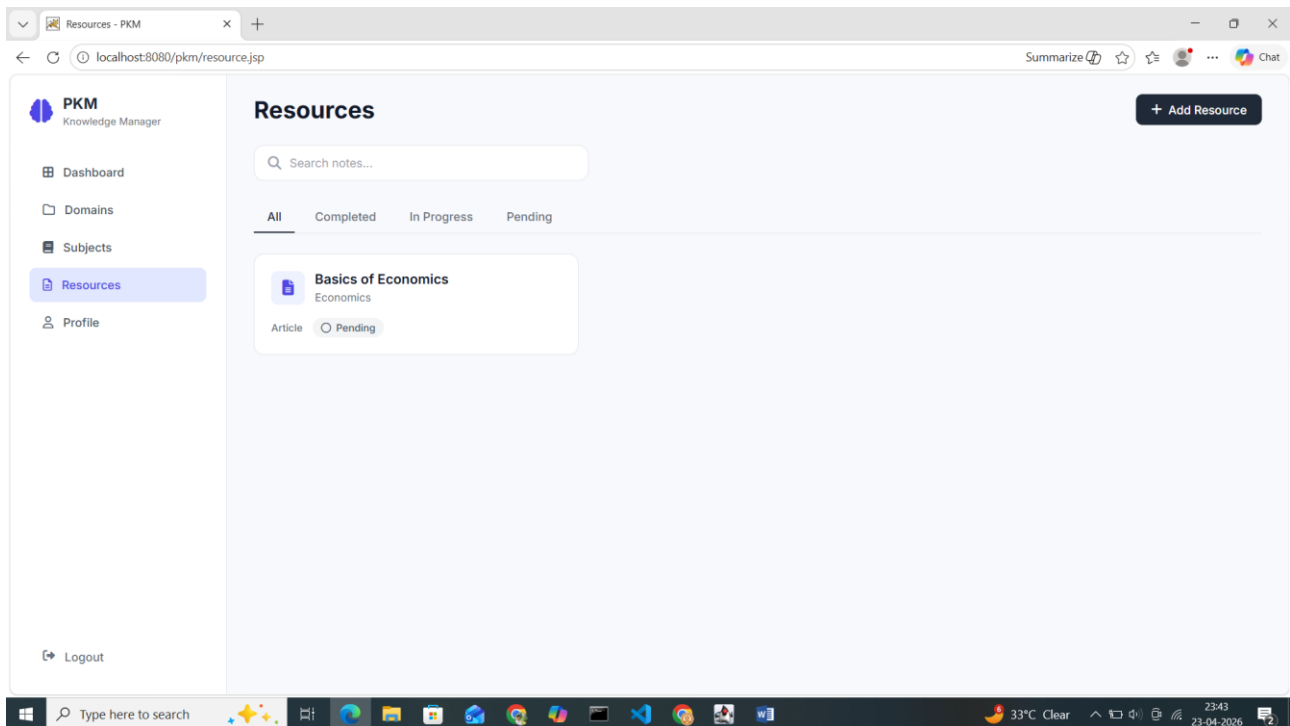


Image10 : Profile Page

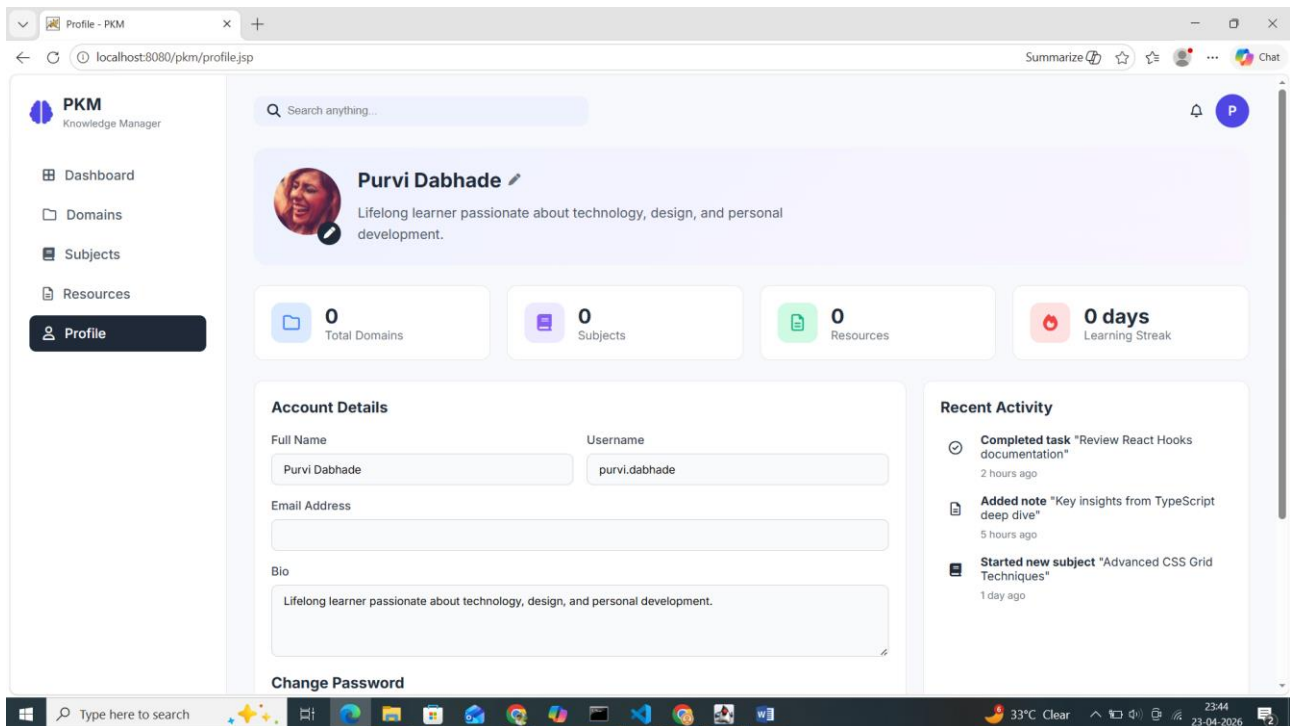


Image11 : Profile Page continue

Profile - PKM

localhost:8080/pkm/profile.jsp

Summarize 0 days Learning Streak

0 Total Domains 0 Subjects 0 Resources

Account Details

Full Name: Purvi Dabhade Username: purvi.dabhade

Email Address:

Bio: Lifelong learner passionate about technology, design, and personal development.

Change Password

Current Password: Enter current password

New Password: Enter new password Confirm Password: Confirm new password

Save Changes

Recent Activity

- Completed task "Review React Hooks documentation" 2 hours ago
- Added note "Key insights from TypeScript deep dive" 5 hours ago
- Started new subject "Advanced CSS Grid Techniques" 1 day ago

Logout

Image12 : Database

```

mysql> desc user;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| user_id | int | NO | PRI | NULL | auto_increment |
| user_name | varchar(25) | YES | | NULL | |
| username | varchar(50) | NO | | NULL | |
| password | varchar(100) | YES | | NULL | |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> desc domain;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| d_no | int | NO | PRI | NULL | auto_increment |
| d_name | varchar(50) | YES | | NULL | |
| user_id | int | YES | | NULL | |
| description | text | YES | | NULL | |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql> desc subject;
ERROR 1146 (42S02): Table 'pkm.subject' doesn't exist
mysql> desc subject;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| s_id | int | NO | PRI | NULL | auto_increment |
| s_name | varchar(50) | YES | | NULL | |
| d_no | int | YES | MUL | NULL | |
| description | text | YES | | NULL | |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> desc resource;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| r_id | int | NO | PRI | NULL | auto_increment |
| r_name | varchar(100) | YES | | NULL | |
| r_type | varchar(50) | YES | | NULL | |
| s_id | int | YES | MUL | NULL | |
| type | varchar(20) | YES | | NULL | |
| link | text | YES | | NULL | |
+----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

```

Image13 : Database continue

```
Command Prompt - "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p
mysql> desc resource;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| r_id  | int  | NO   | PRI | NULL    | auto_increment |
| r_name | varchar(100) | YES |     | NULL    |               |
| r_type | varchar(50) | YES |     | NULL    |               |
| s_id  | int  | YES  | MUL | NULL    |               |
| type  | varchar(20) | YES |     | NULL    |               |
| link  | text  | YES  |     | NULL    |               |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> desc tracking
->
```

FEATURES

- **User Authentication**
Secure login system with session management to ensure authorized access.
- **Interactive Dashboard**
Displays an overview of total domains, subjects, resources, and progress.
- **Domain Management**
Allows users to create and manage different learning domains.
- **Subject Management**
Enables users to add and organize subjects under specific domains.
- **Resource Management**
Allows storage of learning resources such as videos, articles, and notes.
- **Hierarchical Organization**
Maintains a structured flow of Domain → Subject → Resource for better organization.
- **Progress Tracking**
Tracks task status as Completed, Pending, or In Progress.
- **Learning Streak**
Monitors user consistency based on daily activity.
- **Search and Filter Functionality**
Helps users quickly find subjects and resources.
- **User-Friendly Interface**
Provides a clean, responsive, and easy-to-use design.
- **Secure Data Handling**
Uses PreparedStatement and MD5 hashing for secure operations.
- **Dynamic Data Processing**
Supports real-time data insertion, retrieval, and updates using JSP and JDBC.

RESULT

The Personal Knowledge Management (PKM) System was successfully designed and developed as a dynamic web application using JSP, JDBC, HTML, CSS, and JavaScript. The system effectively allows users to manage their learning structure by organizing Domains, Subjects, and Resources in a structured manner.

All core functionalities such as user authentication, session management, data insertion, retrieval, and display are implemented and working correctly. The system enables users to add subjects through an interactive modal interface, view categorized learning content, and track their learning progress.

Database connectivity using JDBC ensures reliable data storage and retrieval, while session handling provides secure access control. The user interface is responsive and user-friendly, enhancing overall usability and experience.

The project successfully demonstrates integration of frontend and backend technologies to build a complete and functional web-based system.

CONCLUSION

The PKM System achieves its objective of simplifying personal learning management by providing a structured and efficient digital platform. It eliminates the need for scattered notes and unorganized resources by centralizing all learning data in one place.

The use of JSP and JDBC enables dynamic data handling, while session management ensures secure and personalized user experiences. Additionally, the implementation of MD5 hashing enhances security by protecting user credentials.

Overall, the project highlights practical application of web development concepts, database integration, and user interface design. It serves as a scalable foundation that can be further enhanced with advanced features such as analytics, reminders, and cloud integration.

The system is reliable, easy to use, and effective for students or individuals aiming to track and improve their learning journey.